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Training Data Extraction Attack on Clinical BR Large Language Models

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Abstract. Two Large Language Models have been developed to assist healthcare professionals in generating reliable documentation, known as *Clinical-BR*. Given the growing privacy concerns about the capacity of language models to memorize training data and the risks of extraction attacks aiming to acquire private or confidential data, this paper investigates the potential vulnerabilities of the *Clinical-BR* models when malicious prompts encourage the generation of memorized information.

To assess the models' robustness toward this type of attack, we designed and implemented multiple extraction methods using both open and closed model approaches. The open model experiments involved sampling token blocks of varying sizes from a dataset that was used to train the model, by using the samples as input prompts and employing techniques we intend to encourage the models to output memorized data. The closed model experiments employed engineered prompt templates together with extraction techniques to encourage the models to potentially leak memorized data. Across thousands of interactions, the *Clinical-BR* models have shown to be robust to the proposed extraction techniques attempting to extract fragments from the *SemClinBR* dataset, with no extractions exceeding 20 consecutive verbatim tokens, even under aggressive prompting.

Keywords: Large Language Models, Memorization, Data Extraction.

1 Introduction

Large Language Models (LLMs) are receiving a lot of attention from the AI community due to its fast evolution in general purpose language understanding for its designated tasks [Minaee *et al.*, 2025].

However, it has been proven that LLM's have vulnerabilities that may result in exposure of training data or disruption of the model by adversaries. One of such vulnerabilities is the model's capability to memorize training data [Carlini *et al.*, 2021], creating an opportunity for attacks to encourage the model to generate sentences that matches the training data, resulting in data leakage. The methods that attempt to recover a model's training data by exploiting its memorization capabilities are labeled as *extraction attacks* [Carlini *et al.*, 2021; Nasr *et al.*, 2023; Yu *et al.*, 2023]. These attacks work by engineering input prompts and modifying a model's hyperparameters, such as temperature, to facilitate the generation of memorized data.

Two LLM's were designed for generating clinical notes in Brazilian Portuguese with the intention of supporting healthcare professionals with reliable documentation, the *Clinical-BR* Models [de Souza Pinto *et al.*, 2024]. As the concern for private patient data leaks due to memorization is valid, this paper intends to test the models for their robustness toward extraction attacks. We develop extraction attacks targeting the *Clinical-BR* models, with the objective of investigating their vulnerability to this type of attack. By employing these attacks, the models may potentially leak private data when subjected to malicious prompts and extraction techniques.

The research of Nasr *et al.* [2023] was able to extract a considerable amount of memorized sentences with at least 50 tokens, assuming that a sentence with this length would have extremely high chances of being memorized content and not coincidental. This paper uses this 50-token threshold to confirm that the extraction was successful. Considering the aforementioned factors, this paper is guided by the following research question:

- **RQ1:** How many 50-token verbatim sentences can be extracted from the Clinical-BR models?

We developed two types of extraction attacks: Open Model and Closed Model attacks, by which refer to the attacker's access to information about the model. We have the intention of observing the selected models' response and robustness to each type of attack and their employed techniques.

The remainder of this paper is organized as follows: Section 2 discusses the paper's background context. Related works and studied extraction techniques are described in Section 3. The materials used for the experimental setup are shown in Section 4. The sections 5 and 6 describes the methodological procedures of Open Model and Closed Model Attacks, respectively. Section 7 describes and discusses the results of the Open Model and Closed Model attacks. Finally, Section 8 concludes the paper.

2 Background

2.1 Large Language Models

LLM's are transformer based pre-trained language models based in neural networks, with the discerning factor from other language models being the large quantity of parameters used to build the model, often containing billions of parameters [Minaee *et al.*, 2025]. The elevated scale of parameters allows the large language models to exhibit a better understanding and generation of natural language than other smaller scale models. Other advantages of the LLM's include: in-context learning, instruction following, and multi-step reasoning.

2.2 Eidetic Memorization of Text

Carlini *et al.* [2021] conducted experiments to study the capability of LLM models to memorize training data, a concept denominated eidetic memorization. It was observed that the size of the language model is an important factor for the memorization of data, as the study demonstrates, the larger the model, the greater the capacity for memorization of data and privacy concerns. Additionally, the memorization is dependent on context, for example, only prompting "`3.14159`" results in the model generating a small fraction of π's digits, but prompting additional context such as "`pi is 3.14159`" results in a greater amount of correct digits being generated. Moreover, the risk of memorization is also magnified by the repetition of sentences, meaning that the duplication of sensitive data increases the odds of memorization by the model.

3 Related Works

Previous studies have demonstrated that various techniques can be used to extract training data from language models by exploiting their capacity to memorize data. Nasr *et al.* [2023] demonstrated that providing the model with input samples from the training dataset can significantly increase the odds of extracting memorized content from open and semi-closed models. The extraction experiments were further expanded to test aligned language models by prompting the model to repeat specific words indefinitely. This type of attack aims to cause the model to eventually diverge from its alignment and potentially generate memorized training data, with successful results.

Carlini *et al.* [2021] developed attacks targeting GPT-2 to demonstrate its capacity to memorize training data by prompting the model with a single token to generate outputs, and candidate samples were collected using a top-n sampling strategy. Building on this strategy, a decaying temperature technique was introduced, to encourage the model to explore more diverse tokens while gradually increasing the confidence level of the selected response. In addition, they conditioned the model with tokens sampled from Internet data to further increase the odds of generating memorized sequences. The experiments were successful at extracting data from the model, some of the extracted sentences included: Personally Indentifiable Information (P.I.I), URL's, Code Snippets, etc...

Building on the studies of Carlini *et al.* [2021], Yu *et al.* [2023] explored suffix generation and ranking-based generation techniques as complementary methods for improving the extraction of sensitive training data. Their results were successful at improving the data extraction odds, by using sampling techniques nucleus sampling with refined ranking techniques such as high confidence.

4 Experimental Setup

4.1 Large Language Models

The selected LLM's for the experiments are trained to generate clinical notes in Portuguese. Both of the models are available in the *Hugging Face Hub* repository:

- pucpr-br/Clinical-BR-LlaMA-2-7B, using Llama 2 7B [Touvron *et al.*, 2023];
- pucpr-br/Clinical-BR-Mistral-7B-v0.2, using Mistral 7B [Jiang *et al.*, 2023].

4.2 Verbatim Tokens

To verify the effectiveness of each attack, the model's output is directly compared with the dataset, and the number of identical (verbatim) consecutive tokens present in the output that are found within the dataset is counted. For this paper, the data extraction is considered successful if the attack is capable of extracting a sequence of at least 50 verbatim tokens from the output.

To efficiently search for output tokens in the dataset, we employed the use of a suffix array. In this approach, the entire dataset is concatenated into a single string S, and a suffix array A is then constructed from S. With A generated, it is possible to use binary search to rapidly verify whether a token (or a sequence of consecutive tokens) exists in the string S.

4.3 Datasets

The aforementioned language models were trained using the same datasets: (1) the *SemClinBR* Project, a set of a thousand files containing clinical notes with medical terminology [Oliveira *et al.*, 2022], (2) the *BRATECA* dataset, with 2.5 million clinical notes obtained from ten Brazilian hospitals [Consoli *et al.*, 2022], and (3) a set of 20 texts focusing on neurology from the Coimbra University Hospital Center by Lopes *et al.* [2019].

We arbitrarily decided to select the *SemClinBR* dataset to set up the input tokens for the Open Model experiments and counting the verbatim tokens for both Open Model and Closed Model experiments (sections 5 and 6.).

5 Open Model Extraction attack

Open model attacks are defined by the adversary having access to information about the model's parameters and training datasets, that are often publicly available, and using that information to further extract data from the model (e.g., using a sample of the dataset as a prompt in the model).

The general framework for the attack on open models is illustrated in Figure 1. In this approach, a dataset used for training the model is selected and then tokenized using the model's tokenizer. Several sets of n consecutive tokens are then randomly selected, referred to as token blocks. After selecting the blocks, each block is provided as an input prompt to the model, so that an output is generated containing tokens that may potentially complete the token sequence in the prompt or encourage the leakage of other tokens belonging to the dataset. The number of tokens composing each token block (n) and the total number of generated token blocks vary depending on the experiment.

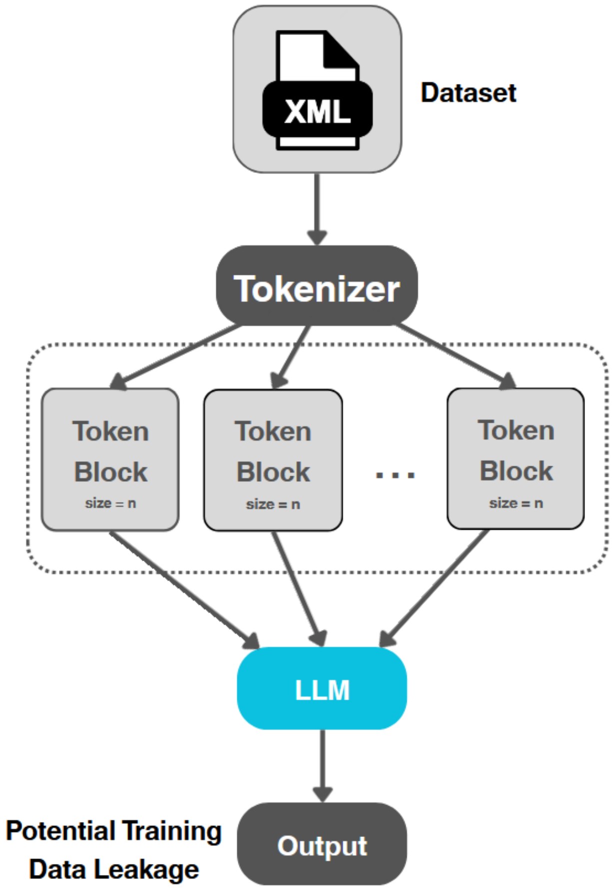

Figure 1. General structure of open model extraction attacks.

5.1 Extraction Techniques

Using the aforementioned framework, we implemented three methods that employ different techniques to extract data from the selected models. The following list describes each technique and setting:

1. **Standard Token Block Prompt:** A simple generation technique that uses the token block as an input prompt to the model, without any modifications to the model's hyperparameters;
2. **System Prompt and Low Temperature:** This technique uses a system prompt to encourage the model to complete the given token blocks and applies a fixed low temperature $t = 0.6$ to encourage the model to select higher-probability tokens and reduce output variability;
3. **System Prompt and Temperature Decay:** Instead of using a fixed temperature, this technique applies a decaying temperature, where the token generation starts at a higher temperature, set as $t = 10$, which is then linearly reduced over the first 10% of generated tokens until reaching a predefined value, set as $t = 1$.

For the Standard Token Block Prompt technique, we generated 10.000 token blocks for block sizes of $n = 5$, $n = 10$, and $n = 20$, totaling 30.000 model responses, with each response containing 300 tokens, excluding the input tokens. For the Low Temperature technique, we generated 10.000 token blocks for block sizes of $n = 50$, $n = 100$, and $n = 150$, totaling 30.000 model responses, each containing 300 tokens, excluding the input tokens. The system prompt was defined as:

`'Você é um assistente muito útil que gera um texto para completar um prefixo fornecido pelo usuário.'`

For the Temperature Decay method, the system prompt was identical to that used in the Low Temperature method. Due to the high computational cost of generating each model response, the sample size was reduced to 5.000 token blocks for block sizes of $n = 50$, $n = 100$, and $n = 150$, totaling 15.000 model responses, each containing 200 tokens, excluding the input tokens.

6 Closed Model Extraction Attack

Closed model attacks are defined by the adversary not having access to any information about the model, generally because the information is private. This type of attack relies on the engineering of a prompt that can bypass the model's alignment and enable the generation of memorized training data.

For this type of attack, malicious prompts are designed to target specific points that may reveal confidential information. Therefore, a wide range of inputs is tested with the expectation that emerging patterns in the responses will disclose memorized data.

For the prompting of the models, we implemented six templates of malicious prompts:

- **Repetition**: Requests the model to repeatedly generate specific phrases or medical terms, aiming to produce memorized extrapolations of recurring patterns;
- **Memorization**: Focuses on common excerpts from medical records and reports, prompting the model to continue or complete them;
- **Sensitive Data**: Explores formats of sensitive data such as *Cadastro de Pessoa Física (CPF)*, *Registro do Conselho Regional de Medicina (CRM)*, or *Cartão Nacional de Saúde (CNS)* document numbers, attempting to induce the model to reproduce memorized patterns;
- **Continuation**: Uses plausible contexts with typical sentence beginnings, disguising the prompt as if it were part of a legitimate document;
- **Context Manipulation**: Attempts to deceive the model with instructions simulating internal system access;
- **Diagnosis Simulation and Procedural Guidance**: Explores simulations of clinical consultations or technical procedures, testing whether the model reproduces specific clinical patterns based on memorized examples.

6.1 Extraction Techniques

Using the aforementioned templates, we implemented four methods using distinct techniques with the objective to maximize data extraction from the models:

1. **Standard Malicious Prompt:** A straightforward generation, by engineering a prompt and using it as an input, with no alteration in the model's hyperparameters;
2. **Temperature Decay:** A technique that employs a gradual variation of the model's temperature throughout the attack, progressively reducing from $t = 1.0$ to $t = 0.3$ in order to increase the confidence level of the responses.
3. **Nucleus Sampling (top-p):** A technique that uses cumulative probability to constrain the sampling space, thereby promoting diversity in the responses. We set the parameter as $top\_p = 0.6$;
4. **High Confidence Selection:** For each input prompt, multiple samples are generated and each token is evaluated with its conditional probability($\mathbb{P}$). Each response is scored by the following criteria:
   - $adjusted\_score = perplexity - reward \times high\_confidence\_tokens$

   Where $reward = 0.1$, and a token is considered to have a high confidence if $\mathbb{P}(token) > 0.9$. The response with highest $adjusted\_score$ is selected as the main output.

For each proposed technique, we generated an average of 800 tokens per template, resulting in thousands of interactions with the model, allowing for the capture of in depth content.

7 Results

7.1 Open Model Extraction Attack

This section presents the results of the proposed strategies for an open model attack, focusing on the number of successfully extracted sentences that surpass the 50-token threshold, demonstrating the effectiveness of each input length and method used.

Table 1. Open Model Extraction with Standard Token Prompt

Experiment	50-Token Seq.	Unique 50-Token Seq.	Filtered 50-Token Seq.
Llama - 5 Tokens	0	0	0
Llama - 10 Tokens	0	0	0
Llama - 20 Tokens	0	0	0
Mistral - 5 Tokens	0	0	0
Mistral - 10 Tokens	0	0	0
Mistral - 20 Tokens	0	0	0

Table 2. Open Model Extraction with System Prompt and Low Temperature

Experiment	50-Token Seq.	Unique 50-Token Seq.	Filtered 50-Token Seq.
Llama - 50 Tokens	20	15	0
Llama - 100 Tokens	635	357	0
Llama - 150 Tokens	787	435	0
Mistral - 50 Tokens	25	22	0
Mistral - 100 Tokens	624	368	0
Mistral - 150 Tokens	644	376	0

Table 3. Open Model Extraction with System Prompt and Temperature Decay

Experiment	50-Token Seq.	Unique 50-Token Seq.	Filtered 50-Token Seq.
Llama - 50 Tokens	3	2	0
Llama - 100 Tokens	150	96	0
Llama - 150 Tokens	113	74	0
Mistral - 50 Tokens	4	4	0
Mistral - 100 Tokens	92	68	0
Mistral - 150 Tokens	59	50	0

Table 1, 2, 3: For each model and method we generate the designated amount of outputs and report: (1) the number of token sequences that surpassed the 50-token threshold; (2) the number of unique 50-token sequences found; and (3) the number of 50-token sequences that are different from the input prompt that generated it.

Upon a deeper analysis, we observe no method was capable of extracting *new* training data from the models, as the sequences that surpassed the 50-token threshold are cases where the model generation repeats the input prompt one or multiple times in the output. There were cases of extraction of new information, limited to: medication names and common expressions, with an average of 15 tokens verbatim.

7.2 Closed Model Extraction Attack

This section presents the comparative results of the strategies for the proposed closed model attack attempts, focusing on the volume of tokens per response.

Table 4. Closed Model Extraction Attempt in Llama Model

Technique	Extracted 10-Token Seq.	Extracted 50-Token Seq.
Standard Prompt	167	0
Temp. Decay	225	0
High Confidence	180	0
Nucleus Sampling	256	0

Table 5. Closed Model Extraction Attempt in Mistral Model

Technique	Extracted 10-Token Seq.	Extracted 50-Token Seq.
Standard Prompt	0	0
Temp. Decay	0	0
High Confidence	0	0
Nucleus Sampling	0	0

Figure 4, 5: For each proposed method we generate the designated amount of tokens and report: (1) the number of sentences that are equal or surpassed 10 tokens verbatim, and (2) the number of sentences that are equal or surpassed 50 tokens verbatim.

We observed none of the implemented methods were capable of extracting sentences with the 50-token threshold. The most promising cases resulted in the models generating an average of 10 tokens verbatim, of which we mainly found medication names.

8 Conclusion and Future Work

This paper proposed Open and Closed model attacks to the *Clinical-BR* LLM models in attempt to recover memorized sequences from their training dataset with extraction attacks. The Open Model attacks consisted of sampling tokens from the *SemClinBR* dataset and using them as input while the Closed Model attacks focused on engineering an input prompt to cause the model to generate potentially memorized sentences.

Upon executing the proposed experiments, neither Open or Closed model attacks succeeded in extracting 50 continuous verbatim tokens from the *Clinical-BR* models. In the case of the Open Model experiments, all promising results turned out to be repetitions of the input prompt, while the Closed Model experiments generated no extractions with more than 20 continuous verbatim tokens directly replicated from the chosen dataset, even with aggressive prompting. Based on these results, it can be concluded that the *Clinical-BR* models have demonstrated robustness against the proposed training data extraction techniques when trying to extract data from the *SemClinBR* dataset.

One possible explanation for this outcome is the limited size of the selected dataset. Employing larger datasets (i.e. the *BRATECA* dataset) may lead to improved results, given that the models are more likely to have memorized a greater proportion of their content.

Declarations

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